PM Rao
Grover

16/1/21

Unit 3
Manufacturing Processes

Raw Material → Manufacturing processes → final product
↓
value addition process
Increasing the value of Raw material

liquid state

Brod classification

- Casting
- Forming
- Fabrication Process
- Metal Removal Process
- Heat treatment process

Primary processes

Secondary Process

Casting
- Sand casting
- Special casting
  - die casting
  - shell molding
  - investment casting
  - centrifugal casting

Force is used
- Rolling
- Extrusion
- Forging
- Sheet metal operation

Forming

Rolling   Extrusion   Forging   Sheet metal operation

Fabrication (joining)

Mechanical joining, Gas welding, Arc welding, Resistance welding
Soldering and Brazing

Metal Removal →
Turning, drilling, shaping, milling, grinding, etc.
Unconventional machining

Mechanical properties

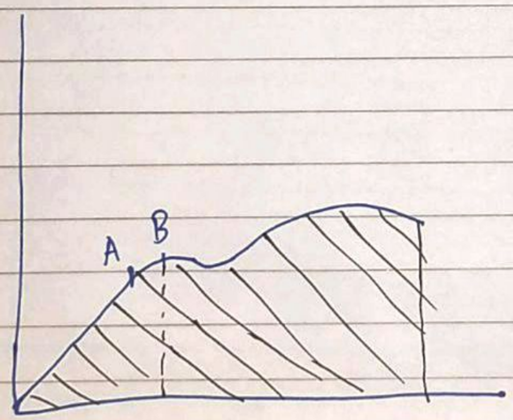
- Strength < Tensile / Compressive

- Toughness - less area, less toughness (strength + ductility)

- Ductility - large deformation before failure (using tensile load)

- Malleability - conversion into sheets/plates using compressive load.

- Brittleness - less deformation before breaking
- Hardness - resistance to deformation
- Stiffness - slope of stress strain curve
- Elasticity - tendency to come back to its original shape within elastic limit
- Plasticity
- Resilience - Energy absorbed upto elastic limit



Energy = Toughness
= area under stress strain curve

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ManuFacturing Processes

Engineering Materials

- Materials Metalic
  - Ferrous
    - Steel
      - Plain Carbon steel
      - Alloy stell
    - Wrought iron
      - ~~Grar~~
    - Cast iron
      - Gray cast iron
      - Malleable
      - Ductile
      -
  - Non-ferrous
- Non-metallic

Non-ferrous
↳ Al, Cu, Mg, Sn, Zn, Pb, Ni etc. and alloy.

Non-metallic

- Organic
  - plastic
  - wood
  - rubber
  - leather
  -
- Inorganic
  - Cement
  - minerals
  - Glass
  - Ceramic
  - Graphite

Mechanical properties : Behaviour of material under force

→ strength → stiffness
→ toughness → Elasticity
→ ductility → Plasticity
→ Brittleness → Resilience - elastic limit
→ Malleability
→ Hardness

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Hardness : resistance

Ductility is permanent deformation before failure.
Plasticity : permanent deformation

Ferrous

Iron ore : Magnetite / Haematite

Steel : Iron + 2% C

Wrough iron 99.5% Fe

Plain carbon steel - Si, P, S, Mn impurities

Steel
- → 0.05 – 0.1 Dead mild steel
- → 0.1 – 0.3 low Carbon steel / Mild steel
- → 0.3 – 0.6 Medium " "
- → 0.6 – 1.7 High " "

decrease ductility ↓

Increase toughness, tensile strength, Hardness

Alloy steel : Si, Cr, Ni, etc.
- Ni → strength and toughness
- Cr → hardness and corrosion resistance
- W → hardness at high temp
- V → tensile strength
- Si → elastic limit increase
- Mn → strength and heat
- Co → hardness
- Mo → extra Tensile strength

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Stainless steel
good corrosion resistance
18/8 steel % of Cr and Ni

High speed steel
W, Mo, Co based steel
18 : 4 : 1 } Best all purpose steel
W Cr V

Cast Iron - % C 2 to 4.5%
→ grey cast iron (free form) Higher comp. strength
→ white cast iron Brittle
→ malleable " " Excellent casting
→ Nodular / Ductile / spheroidal characteristics
Machinability

GCI → graphite flakes
provide lubricant at sliding surfaces
3-3.5 % C 1-2.6% Si 0.15-1% P
0.02 - 0.15% S 0.3 - 0.9% Mn

White Fe_3C Iron carbide / cementite
Hard and Brittle
Combined form 1.7 - 2.4 % C
After Heat treatment 0.2 - 0.85 % P
Annealing 0.85 - 1.2 % Si
0 - 0.12 % S
Wrought Iron can be made

Malleable 2-3 % C 0.6-1.3% Si 0.2-0.6 Mn
upto 0.18% P & S & S &

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find out applications of each type: 445
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Nodular Cast Iron
Mg + → Nodular Cast iron

Most ductile
Used in compressor pumps, agricultural equipments, power transmission

Non Ferrous

1) Aluminium : light weight, Malleable and ductile, corrosion resistant, strength is less than ferrous after making alloy, its application increases.

Duralumin : 4% Cu, 0.5% Mn, Mg and Al
Automobile and aircraft parts
Y- Alloy : 4% Cu, 1.5% Mn, 2% Ni, 6% Si, Mg, Fe, Al
High strength at high temp
aircraft engine parts
Piston
Magnalium : 2-10% Mg ; 1.75 % Cu, Al
- light weight good strength
- aircraft and automobile components

2) Copper : Soft, ductile, malleable, good conductor of heat and electricity.
Brass is a Cu-Zn alloy

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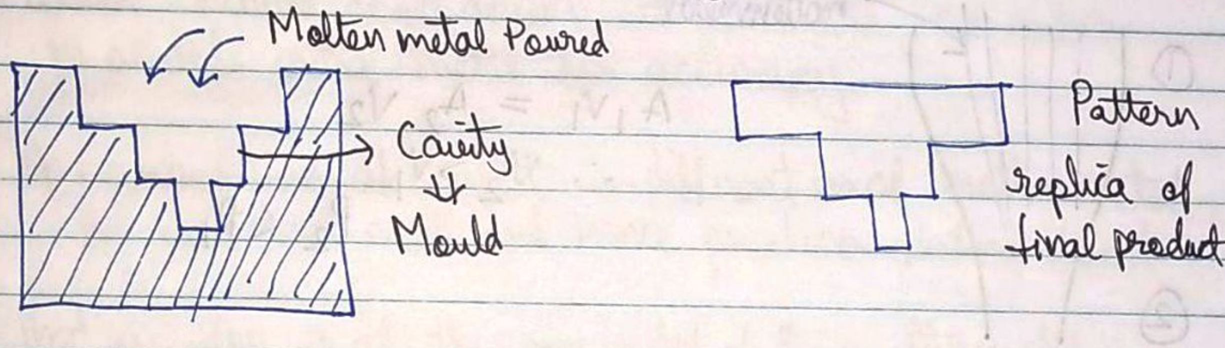
Reinforcement : which provides major strength to material like iron rods in column are reinforcing material

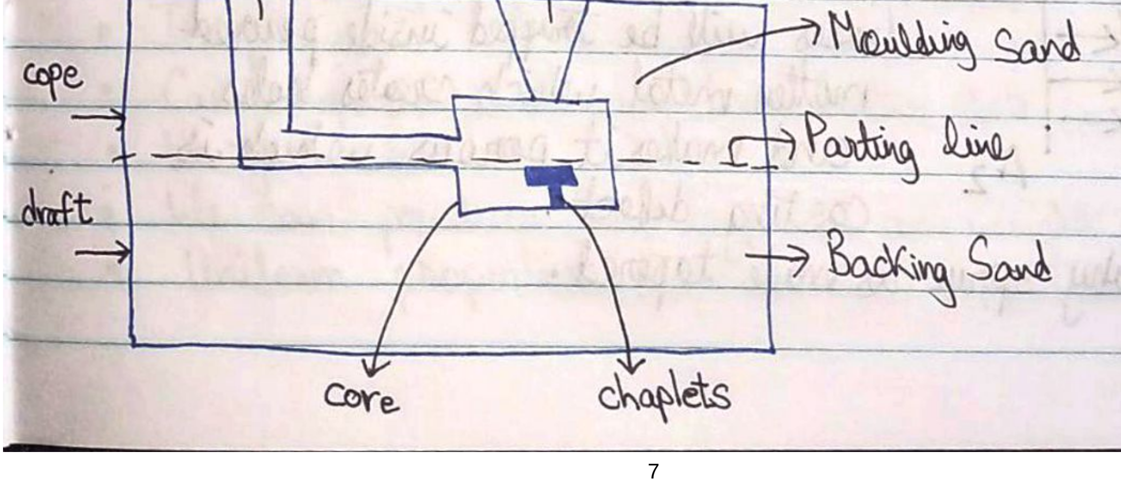
Matrix: Which works as an adhesive
The cement, dust etc. added in column (center)

Manufacturing Processes

1) Casting
→ Pattern and mould
→ Melting and Pouring
→ Solidification and Cooling
→ Removal, cleaning, finishing and Inspection



After solidifying and cooling
remove material, then clean and finishing and inspect



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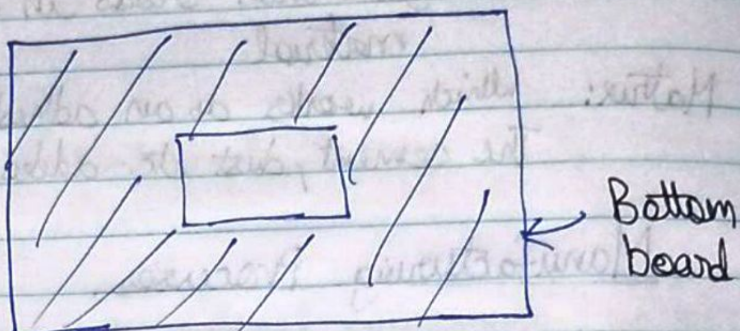
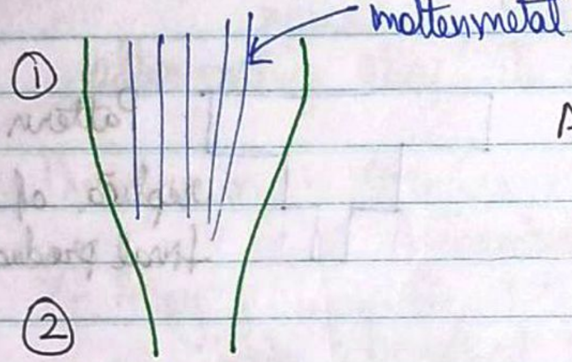
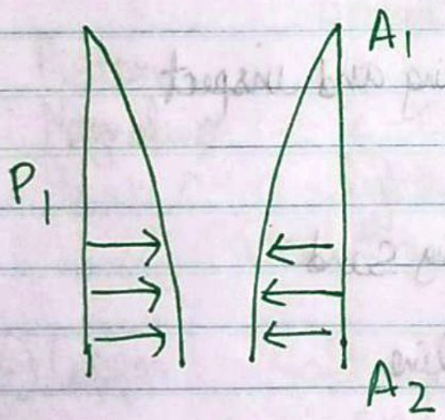
Casting terms

- Flask
- Pattern
- Parting line
- Bottom board
- Facing sand
- Moulding sand
- Backing sand
- Core
- Gating system ⇒ sprue, runner, gate, riser
- Chaplets
- Chills





$$A_1 V_1 = A_2 V_2$$
$$\therefore V_2 > V_1$$
$$P_2 < P_1$$



$P_2 < P_1$

All the surrounding gases will be trapped in section 2.

Gases will be trapped inside poured molten metal which creates holes and makes it porous which is casting defect.

That is why sprue is made tapered.

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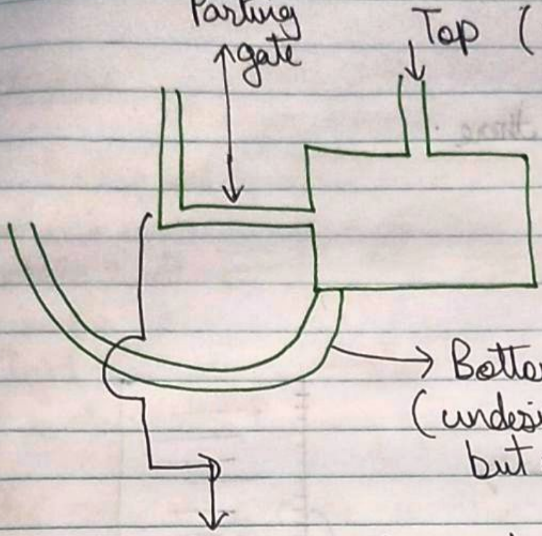
Pouring processes



In riser, metal remains in liquid state for more time. When mould cools down, it contracts, so riser provideds extra metal for accuracy.

If composition of chaplet is different as of molten metal, it will not melt and hence produces casting defect.

That is why is of the same metal of the molten metal.

Advantages of casting

- Can form any complex shape through mould
- Inexpensive
- Cast any material through liquid/molten state.
- Weight reduction
- We can produce any design of any weight.
- Uniform properties (no directional properties)

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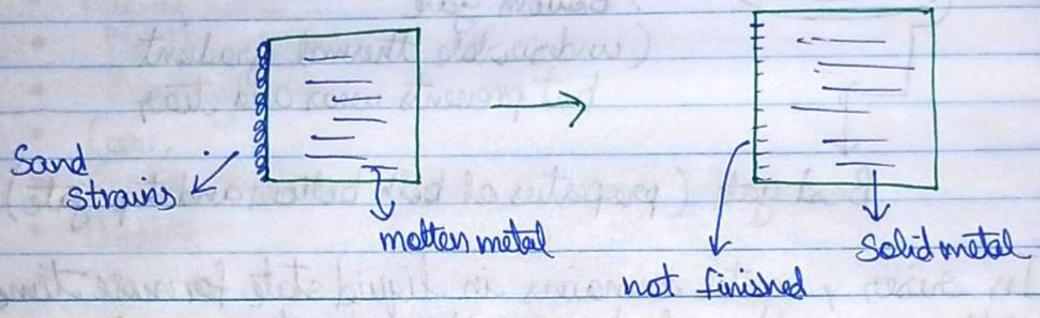
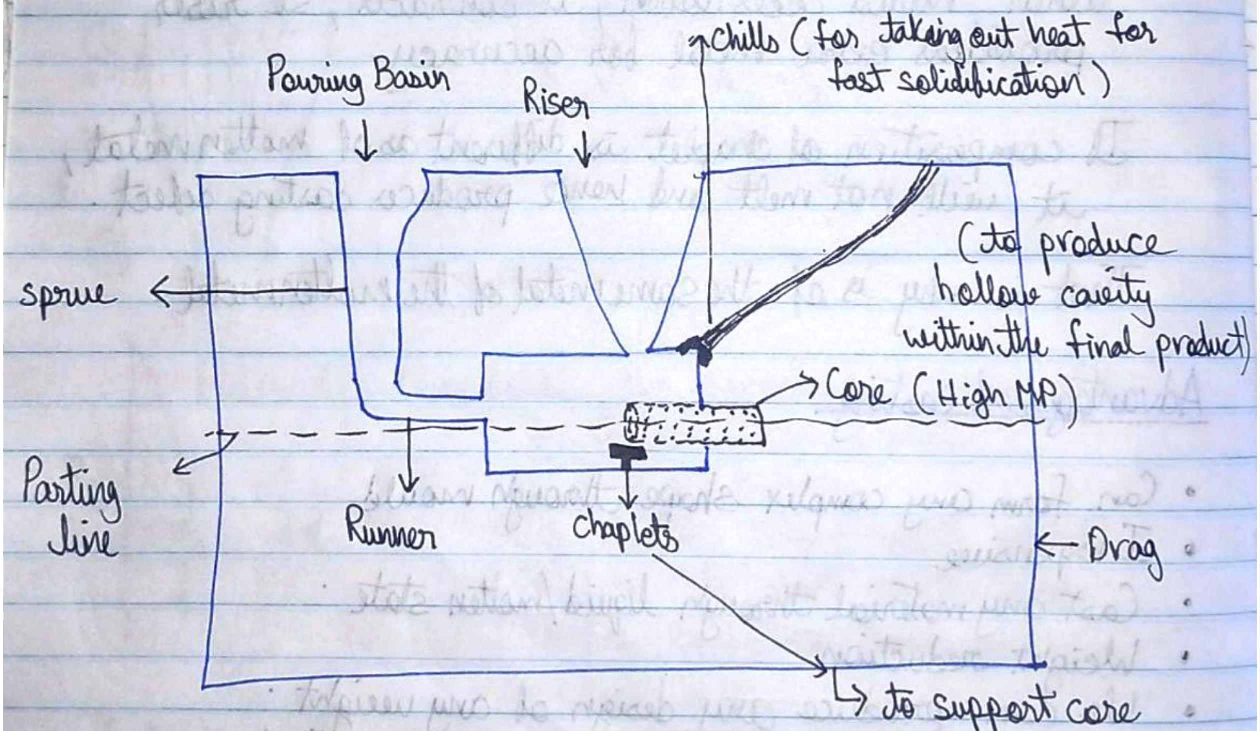
Disadvantages

- Dimensional Accuracy is not there.
- Surface Finish is bad
- Labour Intensive
- Time consuming
- Casting defects





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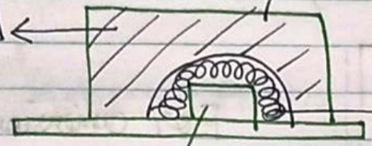
Procedure



It is also called Ramming operation.

In same way cope is also made.
If composition of chaplet is different as of molten metal, it will not melt and hence cause a casting defect.
That is why it is of same metal I.e of molten metal.

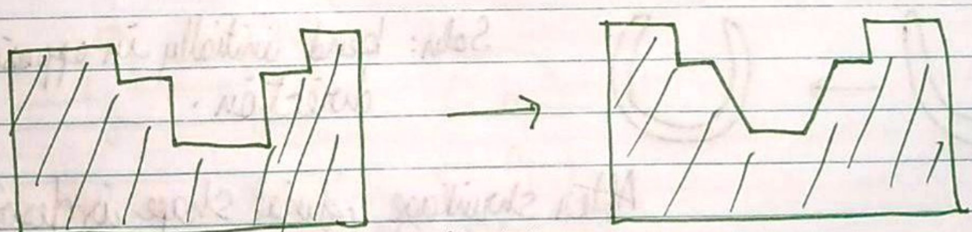
Pattern

↳ Allowance (replica of final product)

- Shrinkage (liquid and solid) → liquid → Solid → Room temp solid (Riser)
- 2 • Draft
- Machining
- Shake
- Distortion

We use increase the pattern size

2) During removal, metal strikes inner walls, wear and tear for easy removal, we make angular.



Draft Allowance

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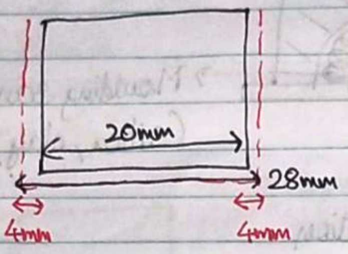
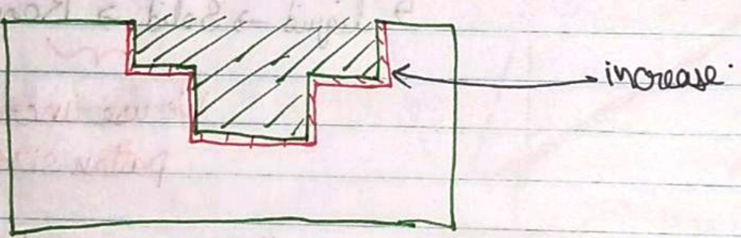
3) Machining : for Surface finish and Accuracy (2-20 mm) we increase the wall.



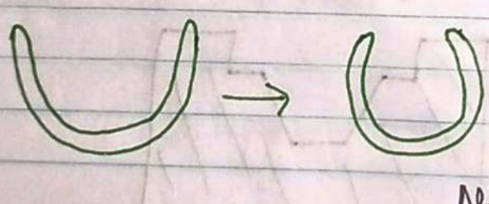
For dimensional accuracy

4) Shake allowance : Property of adhesion of sand particles, they stick to mould wall, which can break/ disturb the mould wall.

Shake, हिलाते the mold, as a result, increase in size of the mold. This is the negative allowance, we initially take smaller pattern.
This is for easy removal.



5) Distortion : if we make thin object, it can shrunk.



Soln: bend initially in opposite direction.

After shrinkage, final shape is desirable

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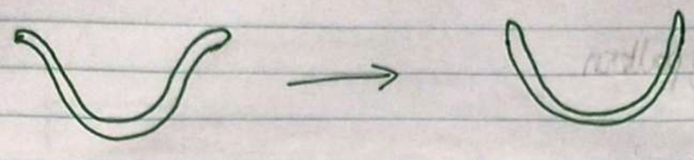


Pattern Material

Wood , Metal , Plastic

Dimensional accuracy is better in Metal rather than Wood.
+1, it has longer durability.
Wood can absorb moisture from moulding sand.

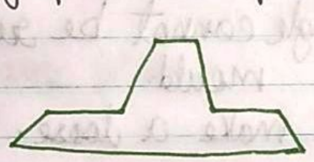
Metal: Cost is high and density is high
See Advantages and Disadvantages from book.
Corrosion property, Density property.

Plastic: lightweight, No corrosion, good surface finish
Sand particles will erode it and life decreases
surface will be disturbed.

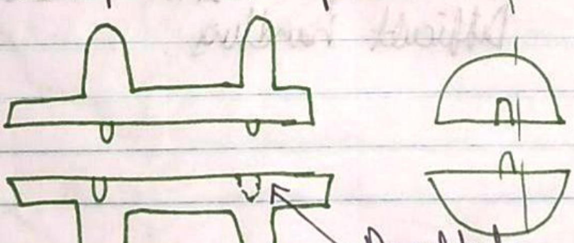
Types of pattern

- Single piece / solid pattern

Simplest



- Two piece / split pattern



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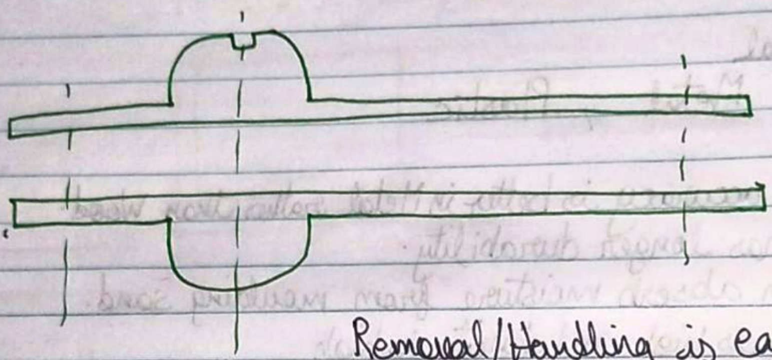
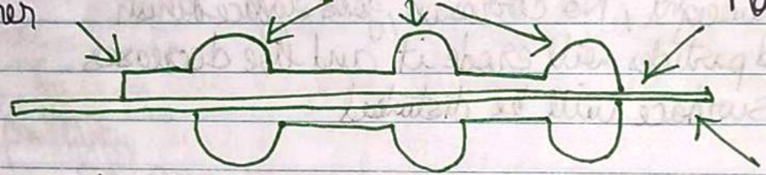
- Cope and drag pattern



Removal/Handling is easy

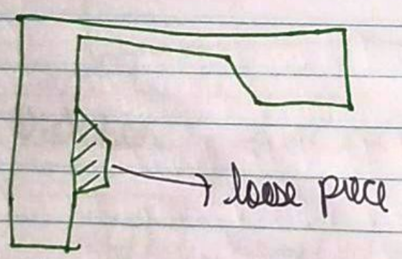
- Match Plate pattern



Productivity ↑

- Loose piece pattern

As a single cannot be removed from mould.
Thus we make a loose.



This is complex and we avoid it.
Difficult handling

- Follow board pattern.

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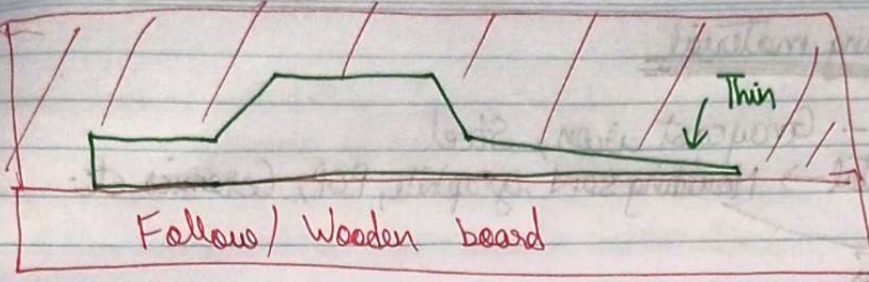
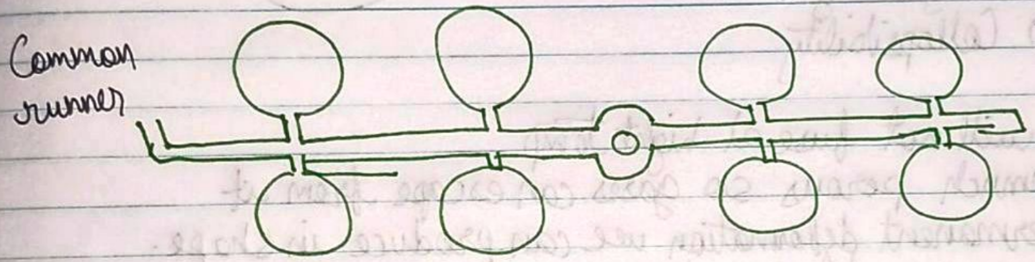


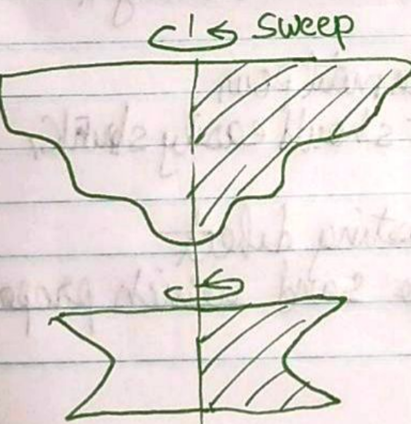
Used when a section is thin, it can deform we use board so it doesnt deform while ramming.

- Gated pattern:
  Multiple gate
  Mass production of casings, multi cavity moulds, Common runner for various patterns.



- Sweep pattern (3D shape) Symmetric Kind

Pivot end is attached.



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Moulding material

Metal – Graycast iron, Steel
Non metal → Moulding sand, graphite, POP, Ceramics etc.

Properties
1 → Refractiveness,
2 → Permeability / Porosity
→ Flowability / Plasticity
→ Adhesiveness
→ Cohesiveness
→ ~~GS~~ Green strength
→ Hot strength
→ Dry strength
→ Collapsibility

Sand

Silica → 70-90%
+
Clay
+
Moisture

1) It will not fuse at hight temp
2) So much porous so gases can escape from it
3) Permanent deformation we can produce in shape.
we can deform it however we want
4) For perfect shape, it should be firm and stick to mould pattern
∴ perfect bonding should be there
5)
6) The mould sand has moisture, at that time strength
8) Without moisture strength
7) Strength at hight temp when molten metal pour
9) Whenever molten metal shrink, it should easily shrink, no resistance to be provided
otherwise stress developed and casting defect
∴ Easy shrinkage of metal allow sand which property is called Collapsibility.

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Cohesiveness : Attraction force between sand particles
property of sand to be firm.

Adhesiveness : bonding between pattern and sand
otherwise there can be deformations.

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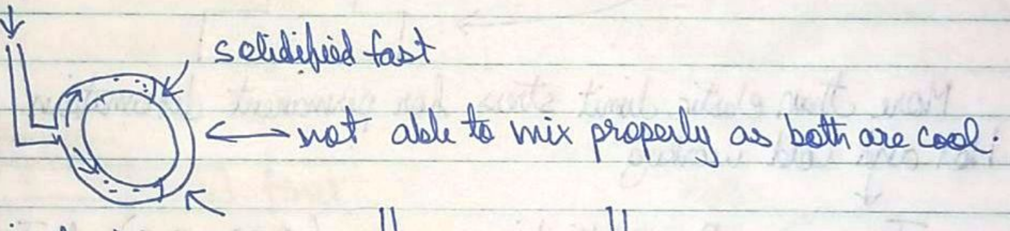
Cohesiveness : Attraction force between sand particles
property of sand to be firm.
Adhesiveness : bonding between pattern and sand
otherwise there can be deformations.

Casting defect

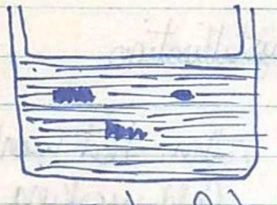
→ gas defect (makes the metal porous, Air inclusion, Blow holes, open h)
→ shrinkage defect
→ Mould sand defect (pattern allowances, defect in shape)
→ Pouring metal defect (Miss run, Cold shut, slag inclusion

solidifies
metal not that hot

Cold shut



Slag inclusion



Mould wall strength, sand टूट के नीचे गिर गया, तो moulding
Sand defect

Special casting process
Shell mould
Permanent die casting - gravity
die casting - pressure
Investment casting
Centrifugal casting

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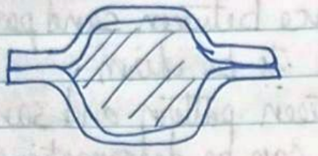
Shell



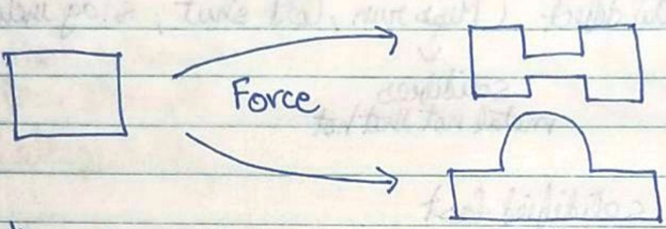
Better surface finish
Dimensional accuracy

Centrifugal -

pipe

Forming Process , Metal working process

one solid form → another solid form

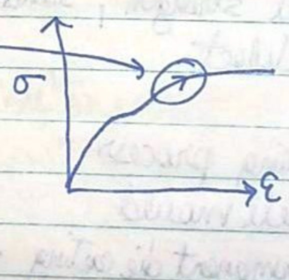


Yield point more

More than elastic limit stress for permanent deformation
Hot and Cold working
↓
Temp , Recrystallisation (0.33 - 0.5) of $T_{melting}$
Cold working

if more than this, then Hot working
if less than this, Cold working

In cold, strain hardening



In hot, no s.h.

In hot, change in dimensions
So after hot working, we use cold working for perfect dimensions.

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Forming Processes

→ Rolling ①

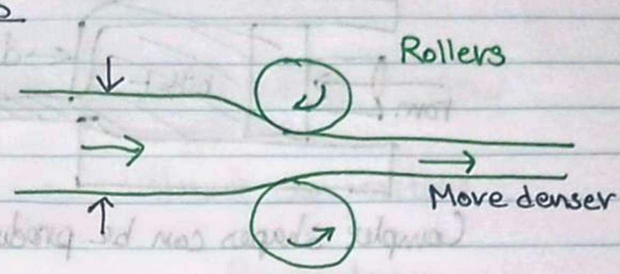
→ Forging ⓘⓘ

→ Extrusion ③

→ Wire drawing

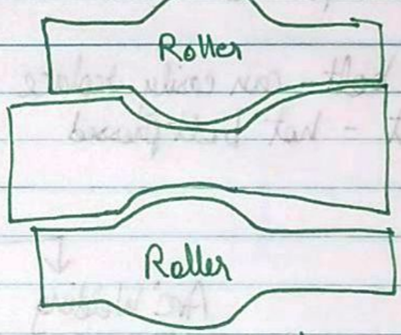
→ Drawing

→ Sheet metal operation

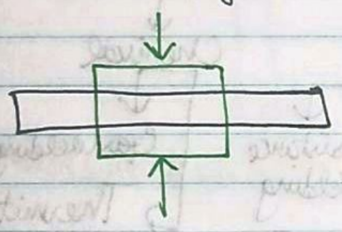
① 

Casting defects, Porosity, impurities are minimized due to compressive force in rollers.

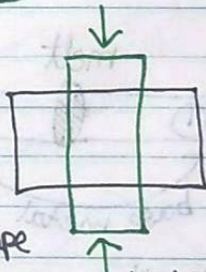
Cover up Profile



(ii) → Drawing out
length ↑ Cross section ↓



Upsetting , length ↓ Cross.S ↑



Smith, Drop, Press, Machine } Forging

- On anvil, Hammer
- Series of blow, simple shape
- Hydraulic, better accuracy, higher shape
- Upsetting, Complex shape, small size

(iii) Extrusion Container , billet

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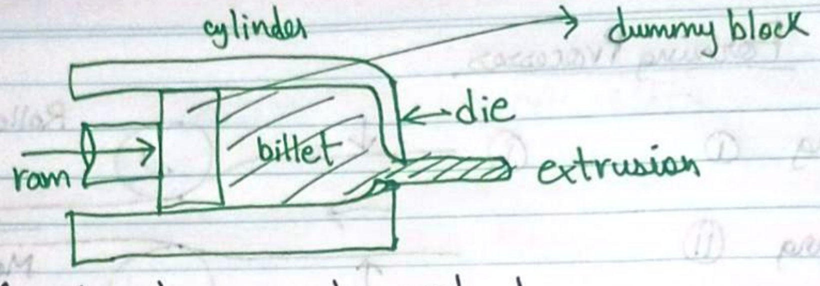


Complex shapes can be produced

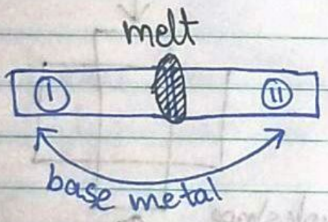
Joining process

Mechanical joining
- Tempory (Nut bolt)
- Semi-permanent (rivets)

Nut bolt - can easily replace
rivet - hot bolt pressed

Atomic bonding (Permanent)
- Solid state (I)
- Liquid state (II)
- Solid / liquid (III)

(II) means
- Electric
  - Arc Welding
  - Induction welding
  - Resistance welding
- Chemical
  - Gas welding
  - Thermit welding



Solid State
- Cold Welding
  - Pressure
  - Explosive
  - USW
- Friction
- Diffusion
- Hot Forged

Solid/liquid
- Soldering
- Brazing
- Adhesive bonding

liquid metal Capillary Action

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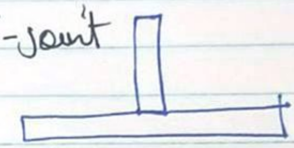
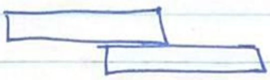
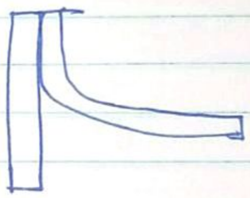
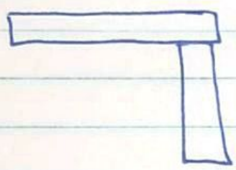
Joint → composition
→ Autogeneous → No filler, no extra material SSW
→ Homogenous → use filler material
→ Heterogenous → filler but different composition

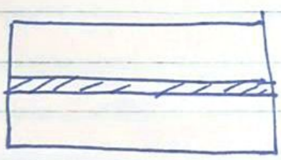
Advantages Disadvantages

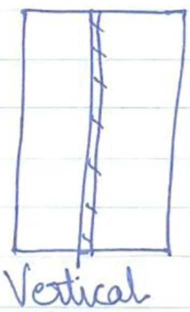
Types of joint

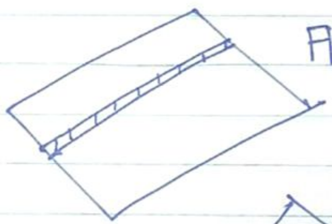
→ Butt Joint

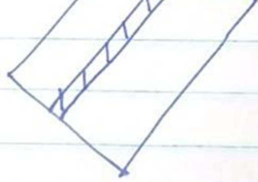
→ T-joint

→ lap Joint

→ Edge joint

→ Corner Joint


Welding positions


Horizontal


Vertical


Flat


Overhead